

Arduino based Traffic light controller

OBJECTIVE: To design a traffic light system which works automatically based on predefined timings.

COMPONENTS USED :

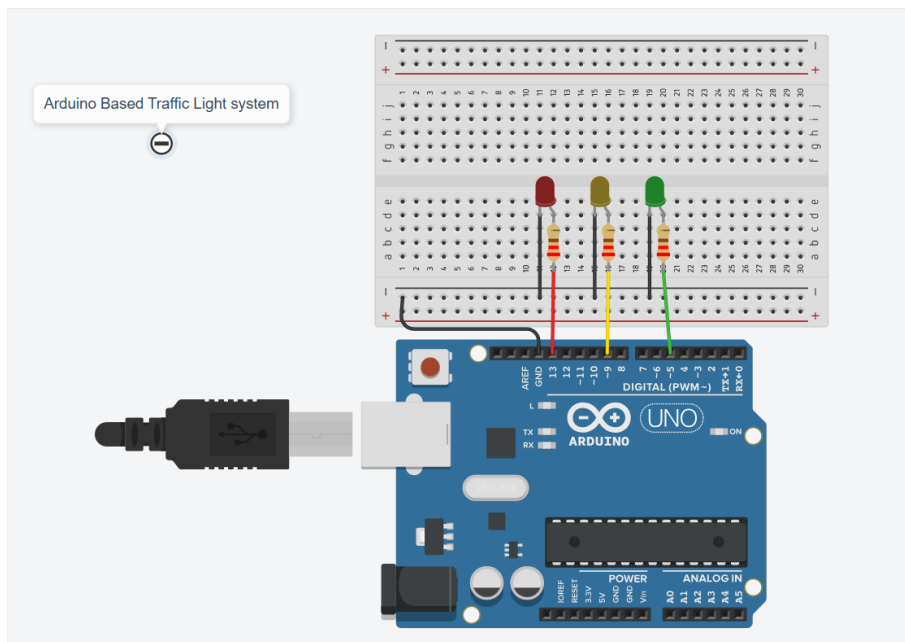
Sno.	Component Used	Count
1.	Arduino UNO	1
2.	Red LED	1
3.	Yellow LED	1
4.	Green LED	1
5.	220 Ω Resistor	3
6.	Breadboard	1
7.	Arduino A-B Cable	1
8.	Male to male Jumper wires	4

PIN CONFIGURATION:

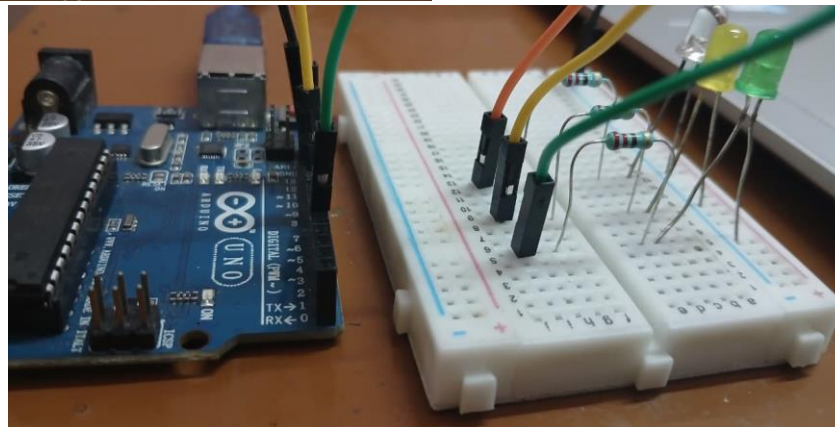
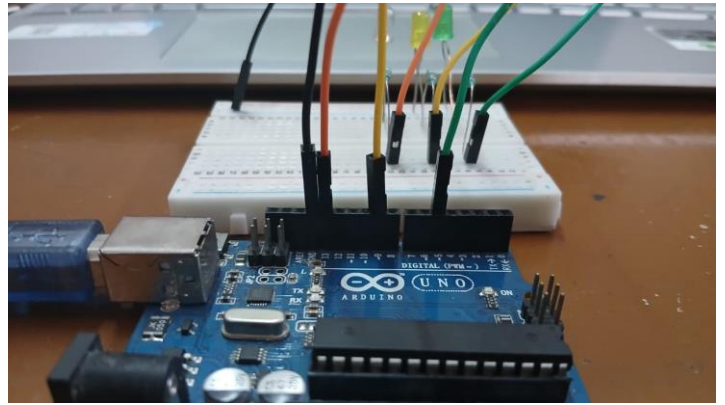
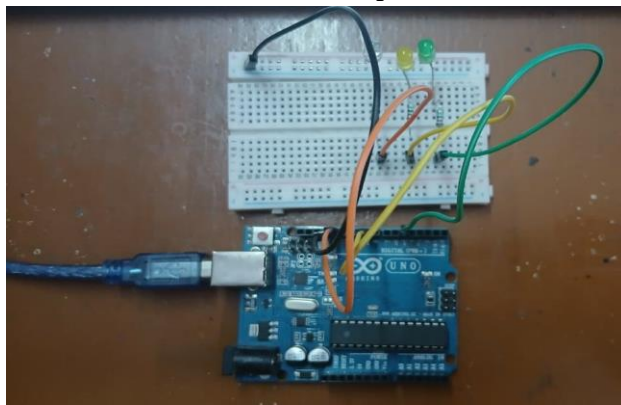
Arduino Pin	Component
D13	Red
D9	Yellow
D5	Green

CIRCUIT DIAGRAM:

1. TinkerCAD circuit:



2. Hardware Circuit Implementation:



THEORY:

- 3 LEDs of red, yellow and green colours placed on the breadboard.
- The longer leg of the LED is the anode (+ve), which is connected to the digital pins
- The shorter leg of the LED, which is the cathode (-ve) is connected to the ground terminal of arduino
- 220 ohms resistors are placed between Arduino digital pins and the leds anode to limit the current going from Arduino to leds.
- It is a control system because the Arduino is the controlling microcontroller, which controls the timing, signal sequence and LED states.

WORKING PRINCIPLE :

- Yellow blinks for 10 seconds, during which traffic can move slowly
- The green light turns on for 10 seconds, during which vehicles can move
- The red light turns on for 10 seconds to stop the traffic.

SOFTWARE CODE & EXPLANATION:

- Pin initialisation:
 pinMode(13, OUTPUT);
 pinMode(9, OUTPUT);
 pinMode(5, OUTPUT);
- Yellow LED blinking loop:
 for (int i=0; i<5; i++){
 digitalWrite(9, HIGH);
 delay(1000);
 digitalWrite(9, LOW);
 delay(1000);
 }
- Green light ON logic:
 digitalWrite(5, HIGH);
 delay(10000);
 digitalWrite(5, LOW);
- Red light ON logic:
 digitalWrite(13, HIGH);
 delay(10000);
 digitalWrite(13, LOW);

```

Traffic_light_system.ino
1  void setup()
2  {
3      pinMode(13, OUTPUT);
4      pinMode(9, OUTPUT);
5      pinMode(5, OUTPUT);
6  }
7
8  void loop()
9  {
10     for (int i=0; i<5; i++){
11         digitalWrite(9, HIGH);
12         delay(1000);
13         digitalWrite(9, LOW);
14         delay(1000);
15     }
16     digitalWrite(5, HIGH);
17     delay(10000);
18     digitalWrite(5, LOW);
19     digitalWrite(13, HIGH);
20     delay(10000);
21     digitalWrite(13, LOW);
22 }
  
```

ALGORITHM:

START

↓

Initialize pins

↓

Blink the yellow LED for 30 sec

↓

Turn ON green LED for 30 sec

↓

Turn ON red LED for 30 sec

↓

Turn OFF all LEDs

↓

STOP

DEBUGGING:

Test_1:

- Yellow led turns on for 10 seconds, but with low brightness
- Green led turns on with full brightness for 10 sec
- Blue LED doesn't turn on

Test_2:

- Replaced the yellow led with another yellow LED and the blue LED with a white led
- Yellow led works fine now
- The white LED doesn't work

Test_3:

- Replaced the white LED with a new red led works fine. The reason for not working earlier: loose wire connection
- All the lights work well

OBSERVATIONS & RESULTS:

1. The yellow light blinks for 5 times
2. The green light turns on for 10 seconds, then turns off
3. Red LED turns on for 10 seconds and then turns off

Test	Result
Yellow LED Blinking	Successful
Green LED timing	Successful
Red LED timing	Successful
Current limiting using a resistor	Successful

Working video: [Hardware demonstration](#)

CHALLENGES FACED:

1. Spark issue on the simulation because of a lack of a resistor
2. Loose wiring due to jumper wires
3. Non-working rusted leds

FUTURE SCOPE :

- Introducing 3 more such systems to make a traffic light system for crossings
- Addition of a pedestrian crossing button.

CONCLUSION:

The project successfully demonstrated a basic traffic light control system using Arduino UNO and LEDs. The simulation verified the proper timing and sequencing of traffic signals.

REFERENCES:

- [TinkerCad Simulation](#)
- [Arduino UNO](#)

